Experiment 1.2: Range and Energy Loss of α-particles in Air

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Report

Abstract

The range and energy loss of α-particles from a ^{241}Am source in air was investigated by measuring the energy spectrum of particles traversing a fixed path length of 64.5 ± 0.5 mm at varying pressures, using a silicon P-N diode detector and multichannel analyser. Three independent methods were applied to determine the stopping pressure: a fixed-exponent power-law approximation of the Bethe-Bloch formula ($k = 0.578$), a floating-exponent variant, and a survival-curve analysis based on the straggling distribution. These yielded ranges in air at atmospheric pressure of 2.80 ± 0.02 cm, 2.52 ± 0.02 cm, and 7.69 ± 0.09 cm respectively, which are in disagreement. The floating-k result is most consistent with the literature value of 5.8 ± 0.1 cm [1], but estimated a significantly lower source energy, indicating a thickness of the gold foil coating on the source being two orders of magnitude above the manufacturer's specification. Straggling analysis revealed that scattering from the gold source coating accounts for $(100 \pm 3)\%$ of the total observed straggling, with air straggling negligible by comparison. No definitive value for the range is concluded; the principal limitations identified are temperature variation between experimental sessions and noise floor contamination of spectra near the stopping pressure.

1 Introduction

α-particles are ionising particles emitted by heavy unstable nuclei. They consist of two protons and two neutrons, making them identical in composition to a helium nucleus. Relative to β-particles and γ-radiation, α-particles have low penetrating power due to their large mass and double positive charge, which drives frequent, intense interactions with surrounding matter. These same properties make α-particles highly damaging to biological tissue: energy is deposited in a highly localised region, causing significant ionisation damage along a short track length.

This ionising character gives α-particles growing relevance in targeted radiative cancer therapy [2], where precise energy deposition is desirable. Understanding and controlling the range of α-particles in air and biological media is therefore essential for developing safe and effective treatments.

A common laboratory source of α-particles is ^{241}Am, a synthetic actinide with a half-life of 432 years, making it a stable and consistent source for experimental use. It decays primarily by α-emission into an excited state of ^{237}Np, with two dominant decay paths emitting α-particles at 5.486 MeV (85.2%) and 5.443 MeV (12.8%) [3], together accounting for 98% of all emissions. The resulting α-particles are stopped by a few centimetres of air at atmospheric pressure.

Energy loss as α-particles traverse a medium arises from stochastic Coulomb interactions with bound electrons and nuclei, and cannot be determined exactly from first principles. The relationships between particle energy, range, and medium properties must therefore be established through theoretical approximations benchmarked against empirical data.

2 Theory

2.1 Bethe-Bloch approximation for range of α-particles in air

The α-particles emitted from the source travel at much less than the speed of light, so the non-relativistic limit is taken. The non-relativistic Bethe-Bloch formula for energy loss of a charged particle of charge z and energy E moving in a medium at velocity v [4] takes the form

$$-\frac{dE}{dx} = A_1 \rho \frac{z^2}{v^2} \ln\left(B_1 v^2\right) \tag{2.1}$$

where A_1 and B_1 are constants dependent on the properties of the medium being traversed[4] and ρ is the density of the material. Considering the non-relativistic relationship between kinetic energy and velocity,

$$E = \frac{1}{2}mv^2 \tag{2.2}$$

where m is the mass of the charged particle, the Bethe-Bloch formula is written in terms of energy as

$$-\frac{dE}{dx} = A_2\rho\frac{1}{E}\ln(B_2E) \tag{2.3}$$

where A_2 and B_2 are constants dependent on the properties of the medium being traversed and the properties of the charged particle. The energy loss of a particle after covering a distance Δx cannot be determined theoretically from this relationship given the discontinuity of the differential equation. Andrews [5] applies a method of linear regression across a relevant energy range of $4-8$ MeV and employs ideal gas law assumptions to approximate the energy E (MeV) of an α-particle of initial energy E_0 after passing through air as

$$E = (E_0^{1.73} - 0.0156\frac{p\Delta x}{T})^{0.578} \tag{2.4}$$

where Δx is the distance traversed (cm), T is the temperature of the air (K) and p is the air pressure (Pa). Consequentially, the stopping pressure, defined as the pressure for which α-particles will be stopped by x_0 cm of air is given as

$$p_0 = 64.31\frac{T}{x_0}E_0^{1.73} \tag{2.5}$$

From this relation, it naturally follows that the range of α-particles in air for a pressure p is given as

$$R = 64.31\frac{T}{p}E_0^{1.73} \tag{2.6}$$

2.2 Energy spectrum of α-particles through air

The stochastic nature of how a charged particle loses energy through traversing air introduces random variation into the range of the α-particles. The measured spread of energy levels detected for a large number of α-particles is expected to follow a skewed Gaussian distribution. A left-skewed form is physically motivated: the maximum detectable energy is set by the source emission energy, creating a hard upper bound, while energy losses prior to detection are unbounded below, generating asymmetry in the low-energy tail. Additionally, the source emits alpha particles at two energies, producing a superposition of two peaks. Given the 51.7 keV separation is small relative to the other modes of broadening, the peaks cannot be resolved, and the skewed Gaussian captures their combined distribution.

It has been shown [6] that the number of particles with the same initial velocity, and by extension the same energy, that the number dn of particles with ranges between x and dx is normally distributed,

$$\frac{dn}{dx} = n_0\frac{1}{\alpha\sqrt{\pi}}\exp(-(\frac{x-R}{\alpha})^2) \tag{2.7}$$

where R is the most probable range, n_0 is the number of initial particles and α is the straggling parameter. The number of surviving particles as a function of pressure is consequentially given by

$$n(p) = \frac{n_0}{2}(1 - erf(\frac{p-p_0}{\alpha_p})) \tag{2.8}$$

where α_p is the pressure straggling parameter, related to α as

$$\frac{\alpha_p}{p_0} = \frac{\alpha}{R} \tag{2.9}$$

The straggling parameter for α-particles is expected to vary linearly with the range as

$$\alpha \approx kR \tag{2.10}$$

where k is the constant of proportionality. In atmospheric pressure the literature value for k is 0.015 [5]. The α-particle source used in this experiment has a gold-foil plating which introduces further straggling. The overall straggling constant from air and the source is given by

$$\alpha^2 = \alpha_{air}^2 + \alpha_{source}^2 \tag{2.11}$$

2.3 Bragg-Kleeman approximation for range of α-particles in other materials

The Bragg-Kleeman rule states that the effective stopping power of materials against ionising radiation is inversely proportional to the square root of atomic mass [7]. This relationship is considered to be accurate to within $\pm 15\%$. Considering a material with density ρ_1 and atomic mass A_1, equation 2.4 is transformed for the energy loss of α-particles through any medium as

$$E = (E_0^{1.73} - 4.464\rho_1 \frac{\sqrt{A_{air}}}{\sqrt{A_1}} \Delta x)^{0.578} \tag{2.12}$$

where E_0 is the initial energy, in this experiment the energy of the α-particles directly after emission and E is the energy of the particles after passing through the medium.

3 Experimental Methods

3.1 Experimental Setup

3.1.1 Apparatus and Particle Path

α-particles were emitted by an ^{241}Am source with a $\sim 3\,\mu$m gold foil coating on the source, which introduced energy loss and scattering. The particles then traversed a fixed distance $x_0 = 64.5 \pm 0.5$ mm through air at a controlled pressure p, accurate within ± 0.005 kPa before passing through a ~ 50 nm gold coating on the detector surface. The thicknesses of both gold coatings are small compared to the uncertainty in x_0.

3.1.2 Detection

A silicon P-N junction diode detector, reverse-biased at 80 V, was used to detect the incident α-particles. The applied bias established a depletion region and reduced the diode capacitance. Incident α-particles generated electron-hole pairs proportional to their energy as they decelerated within the depletion region; these charge carriers were swept to their respective contacts, producing charge pulses proportional to the particle energy.

3.1.3 Signal Processing

The charge pulses were first processed by a charge-sensitive preamplifier, which converted the short charge pulse to a voltage pulse proportional to the deposited charge, with gain set by a feedback capacitor. A shaping amplifier then further amplified and filtered the signal, outputting a $5\,\mu$s Gaussian pulse.

3.1.4 Multi-Channel Analyser

Processed pulses were recorded by a Multi-Channel Analyser (MCA), which accepted input voltages in the range $0 - 1$ V and binned pulse heights into channels, producing a histogram of pulse height against count. A bin count of 1024 was selected as a balance between high energy resolution and low uncertainty. A threshold voltage was applied to suppress low-level noise. The MCA operated on a set live-time interval.

3.2 Data Acquisition

The temperature was measured on the first day of data acquisition to be $24.5 \pm 1.0°C$.

The detection system has inherent noise, centred around the true zero energy of the detector system. A short ($50s$) data acquisition of the noise peak is taken to calibrate the energy readings of α-particle events by bringing the chamber to atmospheric pressure such that all α-particles are blocked from reaching the detector.

The pressure chamber is fully evacuated of air. A 'slow threshold', blocking the 9.1% lowest-energy bins is empirically determined to mask the noise peak. A 100s acquisition time was chosen to yield sufficient counts (approximately 2000 per

bin at peak) for Poisson uncertainties to be small relative to the signal. The pressure is increased in steps of approximately 0.5 kPa. DppMCA fitting software is used to record the approximate mean and FWHM of each peak observed for monitoring throughout data acquisition. The peaks are observed to become more left-skewed as the pressure in the chamber increases. At pressure of 25.81 kPa the noise was noticeably higher than the peak for the α-particle. The slow threshold was increased to 10%. The final reading taken for day 1 was at a pressure of 29.76 kPa.

The temperature was measured on the second day of data acquisition to be $22.7 \pm 1.0°C$. The first measurement taken for this day was at a pressure of 30.05 kPa. A clear negative offset of the central point for this day of approximately 10 bins is observed as a result of the lower temperature. At a pressure of 37.74 kPa. the centre of the peak from α-particle measurements reaches the cutoff point for low energies defined by the 'slow threshold'. A few measurements are taken past this cutoff point, up to a pressure of 39.92 kPa however these measurements have extremely high associated uncertainties.

4 Results

4.1 Energy calibration

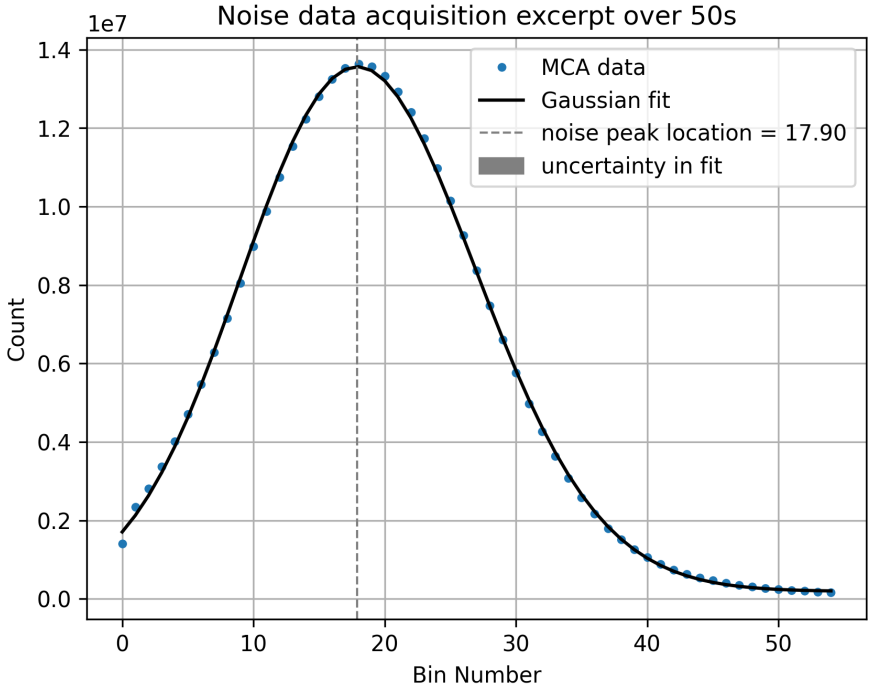


Figure 1: Plot of MCA data for measurement of noise against bin number for a pressure of 101.06 kPa and a live time acquisition of $50s$, fit with a skewed Gaussian (weighted fit from statistical uncertainty). The peak determined by the fitting regime shown as a vertical dotted line at bin number 17.9011 ± 0.0010. Window concentrated around peak for bin numbers $0 - 55$.

A plot of count against bin number is produced for the noise data acquisition in figure 1. The noise peak was fit with the skewed Gaussian function

$$f(x) = \frac{a}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma + b(x-\mu)}\right)^2\right) + c \tag{4.1}$$

where μ is the mean, σ is the standard deviation, a is a scaling constant, c is an offset and b is the skewing factor. The centroid bin number for the noise peak is found to be 17.9011 ± 0.0010.

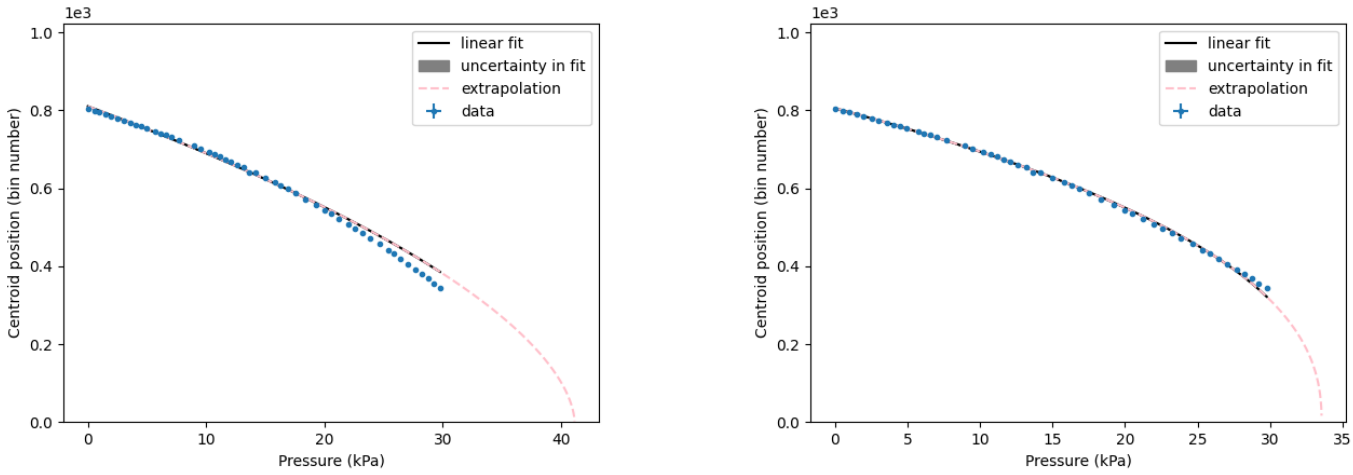
4.2 Energy spectra fitting

A function was developed to process MCA data for α-particle peaks at varying pressures and fit according to equation 4.1 with weighted uncertainties from Poisson statistics. The uncertainty, $u(N)$ relates to the count, N, as

$$u(N) = \sqrt{N} \tag{4.2}$$

The central energy corresponding to each pressure is found as μ. The number of surviving particles was calculated by using the fit parameters to extrapolate the MCA data from the bin for noise offset (18) to bin 1023 and finding the area under the extrapolated graph using a numerical integration regime.

4.3 Centroid energy fit



(a) (b)

Figure 2: The extracted value of centroid position (bin number) from skewed Gaussian fitting of MCA signal plotted against pressure (kPa), corrected with zero energy offset of 17.9011 ± 0.0010.Non-linear fits are in black for a. $k = 0.578$ and b. $k = 0.42075 \pm 0.00007$. Pink dotted lines show the extrapolated fits to x-intercepts of a. 44.136 ± 0.002 kPa and b. 33.550.13. The fit is truer to the data in b. with a reduced χ^2 of 5.2×10^3, a magnitude lower than the χ^2 for a. of 4.6×10^4.

The energy of a particle reaching the detector depends on the pressure of the chamber and the temperature, as described in equation 2.4. It is hence not appropriate to fit the data acquired across both days in order to determine the stopping pressure given the temperature difference observed. Considering that a large proportion of measurements acquired in the second day have high uncertainty due to the overlap of the α-particle peak with the 'slow threshold' mask, the stopping pressure is determined with the data acquired on the first day only. Sufficient data points were acquired to justify this approach.

The relationship between centroid energy and pressure, p is approximated from transforming equation 2.4 to a general form of

$$E = (a - bp)^k \tag{4.3}$$

where a and b are constants, $k = 0.578$ from the approximation by Andrews [5] and E is the centroid bin number, proportional to the expected centroid energy. This method is applied to fit the plot of centroid position against pressure in figure 2a and the fit is extrapolated to find the stopping pressure to be 44.136 ± 0.002 kPa. Using the relation in equation 2.5, the initial energy of the alpha particles is determined as $E_0 = 4.76 \pm 0.02$ MeV. The range of α-particles in air at atmospheric pressure 101.3 kPa is found using equation 2.6 to be 2.80 ± 0.02 cm for the temperature of $297.7 \pm 1.0K$.

A secondary fitting method is applied to determine the stopping pressure, by floating the value of k. This method is applied to fit the plot of centroid position against pressure in figure 2b and the fit is extrapolated to find the stopping pressure to be 33.55 ± 0.13 kPa with a value of $k = 0.42075 \pm 0.00007$. Through transforming equations 2.5, 2.6, replacing 1.73 for the value of $1/k$ found by the fitting regime, the initial energy is calculated as $E_0 = 2.777 \pm 0.011$ MeV and range at atmospheric pressure is found to be 7.69 ± 0.09 cm.

4.4 Gold foil thickness

The thickness of the gold foil coating on the source was calculated using equation 2.12, with the density of gold taken as $\rho_{\mathrm{Au}} = 19.282$ g/cm^3 [8] and atomic mass $A_{\mathrm{Au}} = 196.697$[9]. The emitted α-particle energy used as input was taken as $E_0 = 5.483 \pm 0.003$ MeV, where the central value lies between the most probable emission energy (5.486 MeV [3]) and the intensity-weighted mean (5.480 MeV), with the uncertainty chosen to encompass both. The $\pm 15\%$ model uncertainty of the Bragg-Kleeman rule was propagated in quadrature with the uncertainties in the initial and final energies. Under the fixed-k model, a foil thickness of $\Delta x = 18 \pm 3$ μm was obtained, compared to $\Delta x = (2.0 \pm 0.3) \times 10^2$ μm under the floating-k model.

4.5 Straggling analysis

A plot of the number of surviving particles against pressure is produced in figure 3 and fit according to equation 2.8 to find the stopping pressure to be $p_0 = 39.620 \pm 0.010$ kPa and the pressure straggling parameter to be $\alpha_p = 5.16 \pm 0.09$. Using ideal gas laws, the expected range in atmospheric pressure air corresponding to the stopping pressure is found as $R = 2.52 \pm 0.02$ cm. The range straggling parameter is calculated as $\alpha = 0.840 \pm 0.016$ from equation 2.9. Using the expected value for the straggling parameter in air, given by equation 2.10 as $\alpha_{air} = 0.0378 \pm 0.0003$, is was found that scattering from the gold source accounts for $(100 \pm 3)\%$ of the total observed straggling.

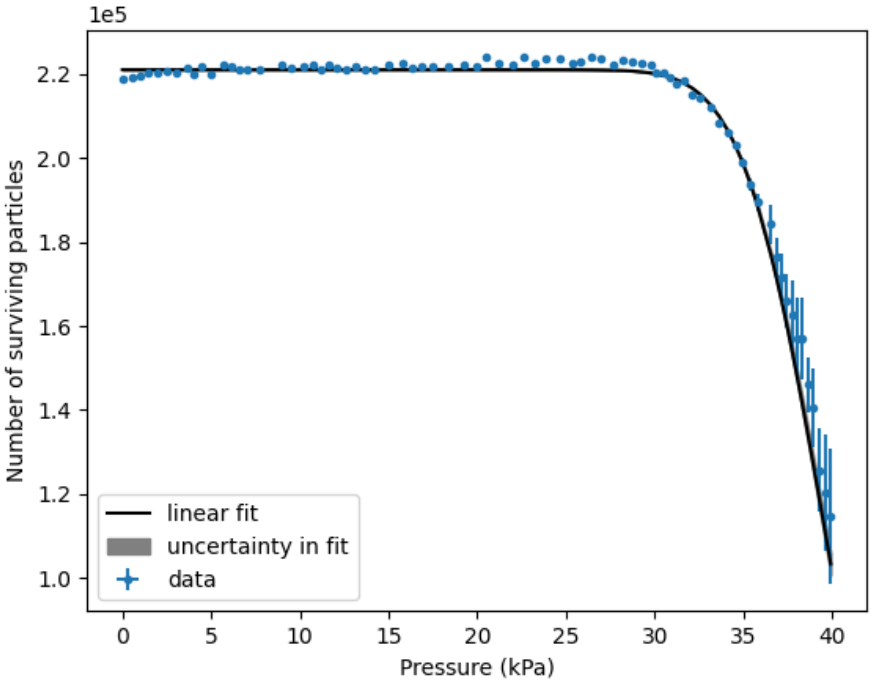


Figure 3: Plot of the number of surviving particles against pressure (kPa). The number of surviving particles is found through integration of the skewed Gaussian functions across a bin range of $18 \rightarrow 1005$ to include all possible energies. A non-linear fit of an error function is shown in black for $p_0 = 39.620 \pm 0.010$, $\alpha_p = 5.16 \pm 0.09$.

4.6 Summary of findings

Table 1: Summary of derived physical quantities from α-particle energy spectra analysis. Method 1 uses a fixed exponent $k = 0.578$ [5] approximation of the Bethe-Bloch law whereas Method 2 floats k as a free parameter.

Quantity	**Method 1 (fixed k)**	**Method 2 (floating k)**
Centroid energy fit		
Exponent k	0.578 (fixed)	0.42075 ± 0.00007
Stopping pressure p_0 (kPa)	44.136 ± 0.002	33.55 ± 0.13
Initial energy E_0 (MeV)	4.76 ± 0.02	2.777 ± 0.011
Range at 101.3 kPa (cm)	2.80 ± 0.02	7.69 ± 0.09
Gold foil thickness Δx	(18 ± 3) μm	$(2.0 \pm 0.3) \times 10^2$ μm
Straggling analysis		
Stopping pressure p_0 (kPa)	39.620 ± 0.010	
Range at 101.3 kPa (cm)	2.52 ± 0.02	
Straggling parameter α (cm)	0.840 ± 0.016	
Expected air straggling α_{air} (cm)	0.0378 ± 0.0003	
Scattering from gold source	$(100 \pm 3)\%$	

Temperature: $T = 297.7 \pm 1.0$ K; Gold density: $\rho_{\text{Au}} = 19.282$ g/cm^3; $A_{\text{Au}} = 196.697$

5 Discussion

The stopping pressure and initial energy of the Am-241 alpha particle source were determined using two fitting methods applied to the centroid energy data acquired on the first experimental day. Fixing the power-law approximated Bethe-Bloch law [4] with an exponent $k = 0.578$ as established by Andrews [5], the stopping pressure was found to be $p_0 = 44.136 \pm 0.002$ kPa, corresponding to an initial energy of $E_0 = 4.76 \pm 0.02$ MeV and a range in air at atmospheric pressure of 2.80 ± 0.02 cm. Floating k as a free parameter yielded a markedly different stopping pressure of $p_0 = 33.55 \pm 0.13$ kPa with $k = 0.42075 \pm 0.00007$, giving $E_0 = 2.777 \pm 0.011$ MeV and a range of 7.69 ± 0.09 cm. A third estimate of the stopping pressure was obtained independently through the straggling analysis, yielding $p_0 = 39.620 \pm 0.010$ kPa and a corresponding range of 2.52 ± 0.02 cm, in closer agreement with the fixed-k result. The gold foil thickness, quoted as 3 μm by the manufacturer, was estimated as $\Delta x = 18 \pm 3$ μm under the fixed-k model, while the floating-k model produced a significantly larger value of $(2.0 \pm 0.3) \times 10^2$ μm.

The straggling analysis revealed that the gold source coating accounts for $(100 \pm 3)\%$ of the total observed straggling, with the contribution from air propagation negligible by comparison. This result is physically reasonable: the gold coating, despite its small thickness, presents a dense, high-Z medium through which α-particles must pass immediately upon emission, producing significant and highly localised energy dispersion before the particles enter the air gap. Air, by contrast, is a low-density medium and the straggling accumulated over 6.45 cm is small relative to that introduced by the coating. The dominance of the source coating implies that the energy spread observed in the MCA spectra is largely an instrumental artefact of the source geometry rather than a property of α-particle propagation in air. This has a practical consequence: the straggling parameter α extracted from the survival curve is not representative of air straggling alone, and comparisons with theoretical predictions for air straggling should account for this. A source with a thinner or absent coating would substantially reduce the total straggling and allow the air contribution to be resolved more cleanly. It would also reduce the skewed Gaussian spread of detected energy values, increasing the accuracy of the median value extracted.

The range of α-particles in air found through the floating-k approximation of the Bethe-Bloch law is the most consistent with the value found by Mohammad et al. [1] of 5.8 ± 0.1 cm at atmospheric pressure (100 kPa), taken as a reasonably good indicator for the range at 1 atm = 101.3 kPa. This value was obtained by varying the distance between the source and the detector at constant pressure. However, the floating-k method indicated an initial energy significantly lower than the expected emission energy of 5.4857 MeV [3], indicating an unrealistically large value for the gold foil thickness, two orders of magnitude higher than the manufacturer's value. The initial energy of $E_0 = 4.76 \pm 0.02$ MeV found by following Andrews [5] is closer to the literature value but still indicates significant energy loss to the gold foil coating, consistent with the significant straggling occurring from the gold foil, however the value for range of α-particles associated with this method is in strong disagreement with the expected value. Given that no method presents itself as significantly more reliable, a final value for the range of α-particles in air cannot be concluded.

The measurements were taken over two days, with the temperature on the second day measured at 2°C/K lower than the first day. A clear negative offset of the mean energy coincides with the data collected on the second day, an observation that

is expected for a decrease in temperature, given the relationship between energy, pressure and temperature in equation 2.4. It was not possible to resolve whether differences in the centroid energy were due to temperature or changing pressures. The approach of only looking at the first day of data addressed this problem by removing temperature as an independent variable, but it also restricted the observation of the energy–pressure relationship at pressures close to the stopping pressure. Additionally, the large variation in temperature across even the first day could have had a significant impact on the collected data. This experiment should be repeated in a temperature-controlled environment, with temperature monitoring throughout the entire data acquisition.

The effect of the temperature differences between the first and second day is not expected to affect calculations of the straggling parameter and stopping pressure from the straggling analysis to the same degree. The interesting component of the plot of surviving particles against pressure in figure 3 occurs at pressures greater than 29.76 kPa, in the second day region. However, it is assumed in the formulation of equation 2.8 that temperature is constant. Given the observed temperature uncertainty of 1°C/K on the second day, this assumption is not entirely valid. A temperature-controlled and monitored experimental setup would improve the validity of these results as well.

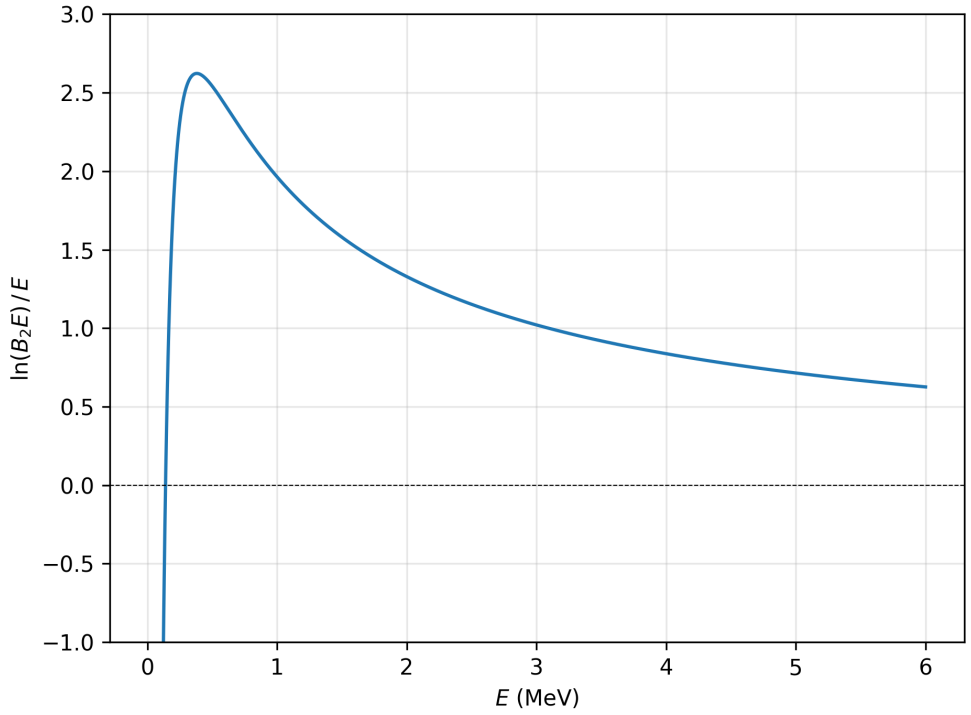


Figure 4: Plot of $\frac{ln(B_2E)}{E}$ against E (MeV) for a value of $B_2 = 7.13\text{MeV}^{-1}$ as calculated according to the Bethe-Bloch [4] formula for air with an electron mass of 9.11×10^{-31} kg, a mean excitation energy of $(10\text{eV}) \times Z$ where $Z = 7.7$[10], an alpha particle mass of $m_\alpha = 6.64 \times 10^{-27}$ kg.

The approximation of the Bethe-Bloch formula as a power law by Andrews [5] is not a reliable method of finding stopping pressure. Observation of the behaviour of the Bethe-Bloch formula 2.3 across a relevant energy range of 0–6 MeV in figure 4 reveals a breakdown of the formula at low energies, where the predicted rate of energy loss becomes negative, which is not physically possible. A breakdown is nonetheless expected, as extremely low energy α-particles behave more like He^{2+} ions and hence interact differently with air, possibly gaining electrons and losing charge. Determining the stopping pressure inherently requires understanding of energy loss processes at low energies as the α-particles progressively slow down. The data collected in this experiment are insufficient to determine the nature of energy loss processes at lower energies. The electrical noise of the detection system prevents analysis of the lowest energies reached by the detector. For further investigation, it would be possible to characterise the relationship between electrical noise and pressure for the detector system, and to estimate and subtract the noise contribution from each MCA spectrum in order to recover the lowest-energy detection events.

A more reliable approach for further investigation into the stopping power of air with respect to α-particles would be to repeat the experiment at varying detector–source separations. A multi-variable relationship between detected energy, pressure, and range would offer deeper insight into the energy loss mechanisms of α-particles travelling through air. Specifically, decreasing the distance between the source and the detector to a few mm would allow data collection at higher pressures, which is more directly relevant to understanding how α-particles behave at atmospheric pressure, and hence to medical applications of α radiation.

6 Conclusion

Three independent methods were used to determine the stopping pressure of ^{241}Am alpha particles in air, yielding ranges of 2.80 ± 0.02 cm and 7.69 ± 0.09 cm for Bethe-Bloch power law approximations and 2.52 ± 0.02 cm with analysis of the straggling distribution. The methods are in systematic disagreement, and no single result can be identified as definitively most reliable; a final value for the range is therefore not concluded. The principal experimental limitations were temperature variation between experimental sessions and noise contamination near the stopping pressure. The experiment was also limited by incomplete and complicated models for the energy loss of charged particles propagating through a medium. Future work should employ a temperature-controlled environment and vary the detector-source separation to better constrain the energy loss relationship across a wider pressure range.

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7 Appendix

7.1 AI Use Statement

In addition to the AI usage for preparation of the logbook, I used Claude/ClaudeCode in the following ways in preparing the work presented here:

1. Production of table 1, through prompting with the results section and asking for a summary table.
2. Writing of the conclusion
3. Error checking